Type MS · Describe what red cells and plasma carry, and from where to where

Answer 1 of 6 · Question 6 part b · 5 marks

2024 JUN · Paper 01 · find part b in this mark scheme extract

6 (a) (i)	3 x 3 x 6; 54 (cm^2)	full marks for correct final answer ecf	(2)
(ii)	3 x 3 x 3; 27 (cm^3)	full marks for correct final answer ecf	(2)
(iii)	54:27; 2:1	ecf for wrong values	(2)
(b)	Any five from • red blood cells contain haemoglobin; • to transport oxygen/binds to oxygen; • blood plasma; • contains dissolved substances; • named substance e.g. nutrients/hormones/CO_2/heat; • transported through blood vessels/named blood vessel;		(5)
(c)	• diffusion; • (nutrients) higher in concentration outside the cell than inside the cell/down a concentration gradient/from high to low concentration;		(2)

Total for Question 6 = 13 marks

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 2 of 6 · Question 1 part a · 1 mark

2020 NOV · Paper 01 (regional R) · find part a in this mark scheme extract

Number				
1	(a)	transport oxygen to body cells		1
	(b)	• (X) plasma (1) • (Y) platelets (1)		2
	(i)	Evidence of division by 5(1) 3:1	full marks for correct final answer	2
	(ii)	• white blood cells/lymphocytes increase in number (1) • (white blood cells)used in defence against disease(1) • produce antibodies/phagocytosis(1)	Ignore fight disease/pathogen Allow kill/destroy pathogen/bacteria	2

Total for Question 1 = 7 marks

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 3 of 6 · Question 1 part a.i · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks
1 (a) (i)	• transport oxygen; • from lungs; • to (body) tissues/cells;	Allow organs	3
(ii)	• haemoglobin to carry/combine with oxygen; • no nucleus, so more haemoglobin; • biconcave shape to increase surface area; • small/flexible (cell) membrane to pass through capillaries; • thin membrane to reduce diffusion distance;		4
(b) (i)	vein;		1
(ii)	• valve present/large lumen/thin (muscular) walls; • only found in veins/not found in arteries/other blood vessels;		2
(iii)	• distance 13-15 mm; • divide by 10 = 1.3-1.5;	Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct	2 Total 12

Question number	Answer	Notes	Marks

Type MS · Describe what red cells and plasma carry, and from where to where

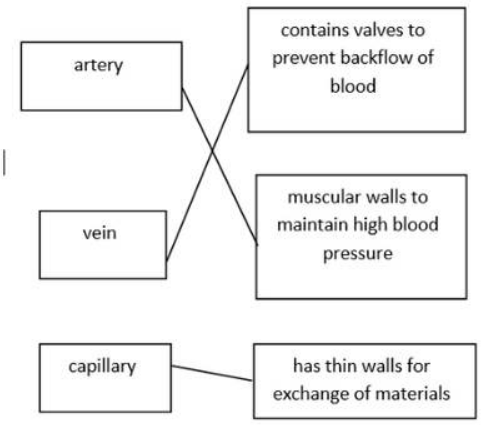
Answer 3 of 6 · Question 1 part a.i · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 4 of 6 · Question 2 part a.ii · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

2 (a) (i)	X = plasma; Y = red blood cell; Z = white blood cell/phagocyte;	Accept named phagocyte	(3)
(a) (ii)	• transports oxygen; • to body/ cells/tissues/organs;		(2)
(a) (iii)	the patient has an infection/leukaemia/allergy/ blood cancer, inflammation;	Accept pathogens in body	(1)
(b)		Reject more than one line from each blood vessel 3 correct=2marks 1 or 2 correct= 1mark	(2)
(c) (i)	left ventricle;		(1)
(c) (ii)	• into right ventricle; • through pulmonary artery; • to the lungs; • (from the lungs) into the pulmonary vein;		(4)

Total for Question 2 = 13 marks

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 5 of 6 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
5 (a)	(i)	• axes labelled (1) • suitable scale (1) • independent variable on X axis (1) • correct plots (2) • suitable curve (1)	-1 for each incorrect plot	6
	(ii)	• mean/average calculated (1) • reliable / reproducible / repeatable (1) • eliminate anomalies (1)		2
(b)	(i)	70 x 100 (1) 90 78%/77.78/77.8 (1)	correct answer = full marks	2
	(ii)	• travels in the blood (1) • time to reach/affect nerve (1)	R heart	2
(c)		• carries impulses to heart muscle (1) • reduces rate of contraction (1)		2

Total for question = 14 marks

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 5 of 6 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 6 of 6 · Question 5 part c · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number			
5 (a)	• addition of figures = 1690; • divide by 10 = 169;		2
(b)	makes it faster/decreases reaction time;		1
(c)	• caffeine has to be absorbed; • into blood; • transported to nervous system/brain;	Ignore body	3
(d)	• can't control what is in food/drink; • could contain caffeine/other factors that would affect result/to ensure caffeine is the cause;	Ignore coffee	2
(e)	185 ± 5 mm;		1 Total 9

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 6 of 6 · Question 5 part c · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks